10TH CLASS MATHEMATICS — NEW SYLLABUS (PECTAA)

# Exercise 1.4 — Real & Imaginary Parts

Unit 1 | Solved Exercise | New Book (PECTAA)

## Exercise 1.4 (Solved)

**1. Find the real and imaginary parts of the following complex numbers:**

(i) $(8 - 3i)^2$

(ii) $(5 + 3i)^{-1}$

(iii) $(4 - 5i)^{-1}$

(iv) $(4 - 3i)^{-2}$

(v) $\left(\frac{3 + 2i}{4 + 3i}\right)^{-1}$

(vi) $\left(\frac{2 - i}{2 + i}\right)^{-2}$

(vii) $\left(\frac{1 - 2i}{1 + i}\right)^{2}$

Write every result in the form a + bi. Then a is the real part and b is the imaginary part.

### (i) $(8 - 3i)^2$

$(8 - 3i)^2 = (8)^2 - 2(8)(3i) + (3i)^2$

$= 64 - 48i + 9i^2$

$= 64 - 48i - 9$

$= 55 - 48i$

**Answer: Re = 55, Im = −48**

### (ii) $(5 + 3i)^{-1}$

$(5 + 3i)^{-1} = \frac{1}{5 + 3i} \times \frac{5 - 3i}{5 - 3i}$

$= \frac{5 - 3i}{(5)^2 - (3i)^2}$

$= \frac{5 - 3i}{25 + 9} = \frac{5 - 3i}{34}$

$= \frac{5}{34} - \frac{3}{34}\,i$

**Answer: Re = 5/34, Im = −3/34**

### (iii) $(4 - 5i)^{-1}$

$(4 - 5i)^{-1} = \frac{1}{4 - 5i} \times \frac{4 + 5i}{4 + 5i}$

$= \frac{4 + 5i}{16 + 25}$

$= \frac{4 + 5i}{41}$

$= \frac{4}{41} + \frac{5}{41}\,i$

**Answer: Re = 4/41, Im = 5/41**

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**(iv) $(4 - 3i)^{-2}$**

$$(4 - 3i)^{-2} = \frac{1}{(4 - 3i)^2}$$

$$(4 - 3i)^2 = 16 - 24i + 9i^2$$

$$= 16 - 24i - 9$$

$$= 7 - 24i$$

$$\therefore \frac{1}{7 - 24i} = \frac{1}{7 - 24i} \times \frac{7 + 24i}{7 + 24i}$$

$$= \frac{7 + 24i}{(7)^2 + (24)^2}$$

$$= \frac{7 + 24i}{49 + 576} = \frac{7 + 24i}{625}$$

$$= \frac{7}{625} + \frac{24}{625}\, i$$

**Answer: Re = 7/625, Im = 24/625**

**(v) $(3 + 2i/4 + 3i)^{-1}$**

$$\left(\frac{3 + 2i}{4 + 3i}\right)^{-1} = \frac{4 + 3i}{3 + 2i}$$

$$= \frac{4 + 3i}{3 + 2i} \times \frac{3 - 2i}{3 - 2i}$$

$$= \frac{12 - 8i + 9i - 6i^2}{9 + 4}$$

$$= \frac{12 + i + 6}{13} = \frac{18 + i}{13}$$

$$= \frac{18}{13} + \frac{1}{13}\, i$$

**Answer: Re = 18/13, Im = 1/13**

**(vi) $(2 - i/2 + i)^{-2}$**

$$\left(\frac{2 - i}{2 + i}\right)^{-2} = \left(\frac{2 + i}{2 - i}\right)^2$$

$$\frac{2 + i}{2 - i} = \frac{2 + i}{2 - i} \times \frac{2 + i}{2 + i}$$

$$= \frac{4 + 4i + i^2}{4 + 1}$$

$$= \frac{3 + 4i}{5}$$

$$\therefore \left(\frac{3 + 4i}{5}\right)^2 = \frac{9 + 24i + 16i^2}{25}$$

$$= \frac{9 + 24i - 16}{25}$$

$$= \frac{-7 + 24i}{25}$$

$$= -\frac{7}{25} + \frac{24}{25}\, i$$

**Answer: Re = −7/25, Im = 24/25**

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**(vii) $(1 - 2i/1 + i)^2$**

$$\frac{1-2i}{1+i} = \frac{1-2i}{1+i} \times \frac{1-i}{1-i}$$

$$= \frac{1-i-2i+2i^2}{1+1}$$

$$= \frac{1-3i-2}{2}$$

$$= \frac{-1-3i}{2}$$

$$\therefore \left(\frac{-1-3i}{2}\right)^2 = \frac{1+6i+9i^2}{4}$$

$$= \frac{1+6i-9}{4}$$

$$= \frac{-8+6i}{4}$$

$$= -2 + \frac{3}{2}\,i$$

**Answer: Re = −2, Im = 3/2**

**2. Solve the following simultaneous linear equations with complex coefficients for w and z:**

(i) $3z + (2 + i)w = 11 - i$

$(2 - i)z - w = -1 + i$

(ii) $2z + (3 + i)w = 9 - i$

$-iz - iw = -1 + i$

(iii) $z - 4w = 3i$

$2z + 3w = 11 - 5i$

(iv) $-iz + w = 3i$

$2z + 3w = 2$

(v) $2z + (3 + i)w = 1$

$-iz + (1 - i)w = 2$

Method: make one equation ready for one unknown (z or w), put its value in the other equation, and solve. Finally check the answer in the second equation.

**(i) $3z + (2 + i)w = 11 - i$ … (1), $(2 - i)z - w = -1 + i$ … (2)**

From (2): $w = (2 - i)z + 1 - i$

Put in (1): $3z + (2 + i)[(2 - i)z + 1 - i] = 11 - i$

$(2 + i)(2 - i) = 4 + 1 = 5$ and $(2 + i)(1 - i) = 2 - 2i + i + 1 = 3 - i$

$3z + 5z + 3 - i = 11 - i$

$8z = 8$

$z = 1$

$w = (2 - i)(1) + 1 - i$

$= 3 - 2i$

**Answer: z = 1, w = 3 − 2i**

**(ii) $2z + (3 + i)w = 9 - i$ … (1), $-iz - iw = -1 + i$ … (2)**

Divide (2) by −i: $z + w = \frac{-1+i}{-i} \times \frac{i}{i}$

$$= \frac{-i+i^2}{1}$$

$$= -1 - i$$

$$\Longrightarrow w = -1 - i - z$$

Put in (1): $2z + (3 + i)(-1 - i - z) = 9 - i$

$(3 + i)(-1 - i) = -3 - 3i - i + 1$

$= -2 - 4i$

$2z - (3 + i)z - 2 - 4i = 9 - i$

$(-1 - i)z = 11 + 3i$

$z = \frac{11 + 3i}{-1 - i}$

$= -\frac{11 + 3i}{1 + i} \times \frac{1 - i}{1 - i}$

$= -\frac{11 - 11i + 3i + 3}{2}$

$= -\frac{14 - 8i}{2}$

$= -7 + 4i$

$w = -1 - i - (-7 + 4i)$

$= 6 - 5i$

**Answer: $z = -7 + 4i$, $w = 6 - 5i$**

**(iii) $z - 4w = 3i$ … (1), $2z + 3w = 11 - 5i$ … (2)**

From (1): $z = 4w + 3i$

Put in (2): $2(4w + 3i) + 3w = 11 - 5i$

$8w + 6i + 3w = 11 - 5i$

$11w = 11 - 11i$

$w = 1 - i$

$z = 4(1 - i) + 3i$

$= 4 - 4i + 3i$

$= 4 - i$

**Answer: $z = 4 - i$, $w = 1 - i$**

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**(iv) $-iz + w = 3i$ … (1), $2z + 3w = 2$ … (2)**

From (1): $w = 3i + iz$

Put in (2): $2z + 3(3i + iz) = 2$

$2z + 9i + 3iz = 2$

$z(2 + 3i) = 2 - 9i$

$z = \frac{2 - 9i}{2 + 3i} \times \frac{2 - 3i}{2 - 3i}$

$= \frac{4 - 6i - 18i + 27i^2}{4 + 9}$

$= \frac{4 - 24i - 27}{13} = \frac{-23 - 24i}{13}$

$= -\frac{23}{13} - \frac{24}{13}\,i$

$w = 3i + iz$

$= 3i + i(-\frac{23}{13} - \frac{24}{13}\,i)$

$= 3i - \frac{23}{13}\,i + \frac{24}{13}$

$= \frac{24}{13} + (3 - \frac{23}{13})i$

$= \frac{24}{13} + \frac{16}{13}\,i$

**Answer: $z = -23/13 - 24/13\,i$, $w = 24/13 + 16/13\,i$**

**(v) $2z + (3 + i)w = 1$ … (1), $-iz + (1 - i)w = 2$ … (2)**

From (2): $iz = (1 - i)w - 2$

Multiply by $-i$: $z = -i[(1 - i)w - 2]$

$= (-i + i^2)w + 2i$

$= (-1 - i)w + 2i$

Put in (1): $2[(-1 - i)w + 2i] + (3 + i)w = 1$

$(-2 - 2i)w + 4i + (3 + i)w = 1$

$(1 - i)w = 1 - 4i$

$w = \frac{1 - 4i}{1 - i} \times \frac{1 + i}{1 + i}$

$= \frac{1 + i - 4i - 4i^2}{1 + 1}$

$= \frac{5 - 3i}{2}$

$w = \frac{5}{2} - \frac{3}{2}\,i$

$z = (-1 - i)(\frac{5 - 3i}{2}) + 2i$

$= \frac{-5 + 3i - 5i + 3i^2}{2} + 2i$

$= \frac{-8 - 2i}{2} + 2i = -4 - i + 2i$

$= -4 + i$

**Answer: $z = -4 + i$, $w = 5/2 - 3/2\,i$**

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